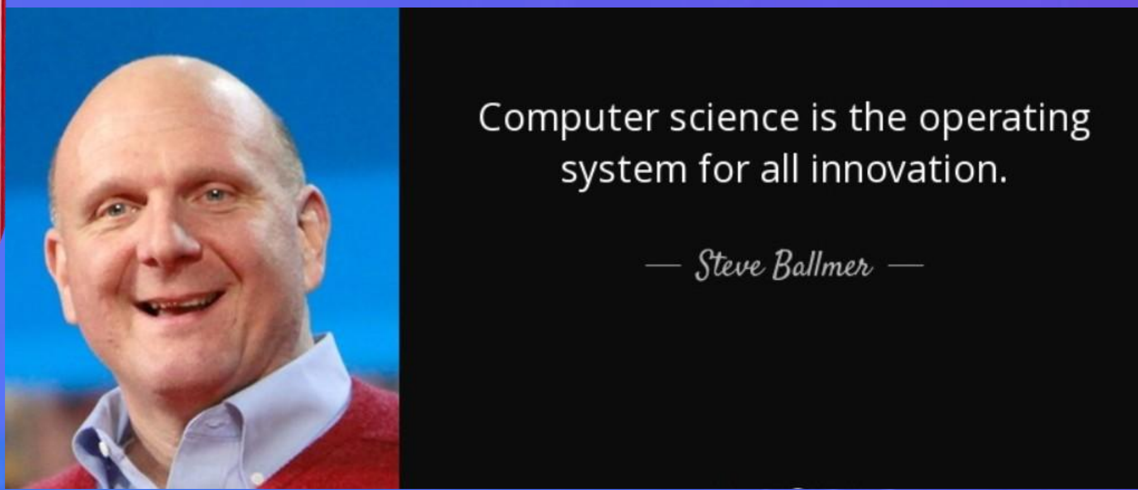
Aeraxia.in

THEORETICAL COMPUTER SCIENCE

CE-SEM 5

VIVA QUESTIONS



Theoretical Computer Science

1. What is Theoretical Computer Science?

Theoretical Computer Science is the mathematical study of computation, its efficiency, and its inherent limitations. It deals with abstract models of computation rather than practical implementation.

2. What is the Chomsky Hierarchy?

The Chomsky Hierarchy is a classification of formal grammars and the languages they generate into four types (Type-0 to Type-3) based on their increasing restrictions and expressive power.

3. Differentiate between a Finite Automaton and a Pushdown Automaton.

A Finite Automaton has finite memory and recognizes Regular Languages. A Pushdown Automaton has an infinite stack as memory and recognizes Context-Free Languages.

4. What is a Regular Expression?

A Regular Expression is a sequence of characters that defines a search pattern, specifically for matching strings in Regular Languages. It uses operators like union (|), concatenation, and Kleene star (*).

5. What is the Pumping Lemma for Regular Languages?

It is a theorem used to prove that a given language is not regular. It states that for any regular language, there exists a pumping length 'p' such that any string longer than 'p' can be "pumped" (its middle section repeated) to generate new strings that must also be in the language.

6. Define a Context-Free Grammar (CFG).

A CFG is a set of recursive rewriting rules (productions) used to generate patterns of strings. It is defined by a 4-tuple (V, T, P, S), where V is variables, T is terminals, P is productions, and S is the start symbol.

7. What is a Turing Machine?

A Turing Machine is a theoretical model of computation that consists of an infinite tape, a head that reads/writes symbols, and a state register. It is capable of simulating any computer algorithm and is used to define the limits of computability.

8. What is the Halting Problem?

The Halting Problem is the undecidable problem of determining, from a description of an arbitrary program and its input, whether the program will finish running or continue to run forever. Alan Turing proved that a general algorithm to solve this problem for all possible program-input pairs cannot exist.

9. What does P vs NP mean?

P is the class of problems solvable in polynomial time by a deterministic Turing Machine. NP is the class of problems verifiable in polynomial time. The P vs NP question asks whether every problem that can be verified quickly can also be solved quickly.

10. What is an NP-Complete problem?

A problem is NP-Complete if it is in NP and every problem in NP can be reduced to it in polynomial time. If a polynomial-time solution is found for any NP-Complete problem, then P = NP.

11. Give an example of an NP-Complete problem.

The Boolean Satisfiability Problem (SAT) is the first problem proven to be NP-Complete.

12. What is a Decidable Problem?

A problem is decidable if there exists a Turing Machine that will always halt and give a "yes" or "no" answer for every input instance of the problem.

13. What is Recursion Theorem?

The Recursion Theorem in computability theory states that a Turing Machine can obtain its own description and then use it in computation. This is central to proofs about self-replication and the halting problem.

14. Differentiate between Moore and Mealy Machines.

Moore Machines' output depends only on the present state, while Mealy Machines' output depends on both the present state and the current input.

15. What is a Universal Turing Machine?

A Universal Turing Machine (UTM) is a Turing Machine that can simulate an arbitrary Turing Machine on arbitrary input. It is a model of a general-purpose computer.

16. What is the Church-Turing Thesis?

It is a hypothesis that any function that is effectively computable can be computed by a Turing Machine. It links the informal concept of an algorithm with the formal model of a Turing Machine.

17. What is a Regular Language?

A language is regular if it can be generated by a Regular Grammar (Type-3) and recognized by a Finite Automaton (DFA/NFA).

18. What is the role of ε (epsilon) in Automata?

Epsilon (ε) represents an empty string or a transition that occurs without consuming any input symbol. It is used in NFA with ε-moves and in certain grammar productions.

19. What is a Deterministic Finite Automaton (DFA)?

A DFA is a finite state machine where for each state and input symbol, the next state is uniquely determined. It does not allow ε-transitions.

20. What is a Non-Deterministic Finite Automaton (NFA)?

An NFA is a finite state machine where, for a given state and input symbol, there can be multiple possible next states. It can also have ε-transitions.

21. Can every NFA be converted to a DFA?

Yes, every NFA can be converted to an equivalent DFA using the subset construction algorithm, though the number of states may increase exponentially.

22. What is a Pushdown Automaton (PDA)?

A PDA is a finite automaton equipped with an infinite stack. It is used to recognize Context-Free Languages.

23. What is Greibach Normal Form?

A CFG is in Greibach Normal Form if every production is of the form $A \rightarrow a\alpha$, where 'a' is a terminal and 'α' is a string of zero or more variables.

24. What is the Pumping Lemma for Context-Free Languages?

It is a theorem used to prove that a given language is not context-free. It states that for any CFL, there exists a pumping length such that certain long strings can be decomposed and "pumped" to generate new strings that must also be in the language.

25. What is an Unrestricted Grammar?

An Unrestricted Grammar (Type-0) has production rules with no restrictions, allowing it to generate recursively enumerable languages, which are recognized by a Turing Machine.

26. What is a Context-Sensitive Grammar (CSG)?

A CSG (Type-1) has production rules where the left-hand side can be replaced by the right-hand side only if it appears in a valid "context". Its languages are recognized by a Linear Bounded Automaton (LBA).

27. Define a Linear Bounded Automaton (LBA).

An LBA is a restricted Turing Machine whose tape length is a linear function of the length of the input string. It recognizes Context-Sensitive Languages.

28. What is a Multitape Turing Machine?

A Multitape Turing Machine has multiple tapes, each with its own read-write head. It is more convenient for certain problems but is computationally equivalent to a standard single-tape Turing Machine.

29. What is Reducibility?

Reducibility is a method for solving a problem by transforming it into another problem that is already known how to solve. If problem A can be reduced to problem B, and B is decidable, then A is also decidable.

30. What is the Post Correspondence Problem (PCP)?

The PCP is an undecidable problem that asks, given a set of dominoes (each with a top and bottom string), whether there is a sequence of these dominoes such that the concatenated top string equals the concatenated bottom string.

31. What is a Recursively Enumerable Language?

A language is recursively enumerable if it is accepted by a Turing Machine. The machine will halt for all strings in the language but may loop forever for strings not in the language.

32. What is a Recursive Language?

A language is recursive (or decidable) if there exists a Turing Machine that halts for every input string and correctly decides whether the string is in the language or not.

33. Differentiate between Decidable and Undecidable Problems.

A decidable problem has an algorithm (Turing Machine) that always gives a correct yes/no answer. An undecidable problem has no such algorithm for all input cases.

34. What is the Complexity Class NP-Hard?

A problem is NP-Hard if every problem in NP can be reduced to it in polynomial time. An NP-Hard problem does not need to be in NP itself.

35. What is a Clique problem?

The Clique problem is to find a complete subgraph of a given size in a graph. It is a classic NP-Complete problem.

36. What is a Hamiltonian Path?

A Hamiltonian Path is a path in a graph that visits each vertex exactly once. Determining whether such a path exists is an NP-Complete problem.

37. What is the Traveling Salesman Problem (TSP)?

The TSP is the problem of finding the shortest possible route that visits each city exactly once and returns to the origin city. Its decision version is NP-Complete.

38. What is a Vertex Cover?

A Vertex Cover of a graph is a set of vertices such that every edge of the graph is incident to at least one vertex in the set. Finding a minimum vertex cover is an NP-Hard problem.

39. What is the Cook-Levin Theorem?

The Cook-Levin Theorem states that the Boolean Satisfiability Problem (SAT) is NP-Complete. This was the first problem proven to be NP-Complete.

40. What is a Regular Grammar?

A Regular Grammar (Type-3) is a formal grammar where every production rule is restricted to a single non-terminal on the left-hand side and a terminal followed by a non-terminal, or just a terminal, on the right-hand side.

41. What is the Arden's Theorem?

Arden's Theorem provides a method to solve a system of equations for a regular expression, specifically for converting a Finite Automaton to a Regular Expression.

42. What is an Epsilon Closure?

The epsilon closure of a state is the set of all states reachable from that state using only ε-transitions.

43. What is a Minimal DFA?

A Minimal DFA for a regular language is a Deterministic Finite Automaton with the minimum number of states possible. It is unique for a given language (up to isomorphism).

44. What is a Myhill-Nerode Theorem?

The Myhill-Nerode Theorem provides a necessary and sufficient condition for a language to be regular, based on the number of equivalence classes of a specific relation defined by the language. It is also used to minimize DFAs.

45. What is a Chomsky Normal Form?

A CFG is in Chomsky Normal Form (CNF) if every production is of the form A → BC or A → a, where A, B, C are non-terminals and 'a' is a terminal.

46. What is a CYK Algorithm?

The Cocke-Younger-Kasami (CYK) algorithm is a parsing algorithm for context-free grammars in Chomsky Normal Form. It uses dynamic programming to check if a string can be generated by the grammar.

47. What is a Lambda Calculus?

Lambda Calculus is a formal system in mathematical logic for expressing computation based on function abstraction and application using variable binding and substitution. It is equivalent to Turing Machines.

48. What is a Primitive Recursive Function?

Primitive Recursive Functions are a class of total computable functions that are defined using basic functions (zero, successor, projection) and operations like composition and primitive recursion.

49. What is a μ-Recursive Function?

μ-Recursive Functions are a class of partial computable functions that extend primitive recursive functions by adding the μ-operator (unbounded search). This class is equivalent to the functions computable by a Turing Machine.

50. What is the Rice's Theorem?

Rice's Theorem states that all non-trivial, semantic properties of programs (i.e., properties of the function the program computes) are undecidable.